JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY, ANANTAPUR

COLLEGE OF ENGINEERING (AUTONOMOUS)ANANTHAPURAMU –515 002 (A.P) INDIA

I B. TECH (R23) – II SEMESTER

Subject Code	Title of the Subject	L	T	P	C
23A20502	Data Structures	3	0	0	3

Course Objectives:

- Toprovidetheknowledgeofbasicdatastructuresandtheirimplementations.
- To understand importance of data structures in context of writing efficient programs.
- Todevelopskillstoapplyappropriatedatastructuresinproblemsolving.

Course Outcomes: At the end of the course, Student will be able to

CO1: Explain the role of linear data structures in organizing and accessing data efficiently in algorithms.

CO2: Design, implement, and apply linked lists for dynamic data storage, demonstrating understanding of memory allocation.

CO3: Develop programs using stacks to handle recursive algorithms, manage program states, and solve related problems.

CO4: Apply queue-based algorithms for efficient task scheduling and breadth-first traversal in graphs and distinguish between deques and priority queues, and apply the map appropriately to solve data management challenges.

CO5: Devise novel solutions to small scale programming challenges involving data structures such as stacks, queues, Trees.

CO6: Recognize scenarios where hashing is advantageous, and design hash-based solutions for specific problems.

	POI	PO2	P03	PO 4	PO5	PO6	PO7	PO8	PO 9	POI O	POI I	POI 2	PSO I	PS0 2	PS0 3
CO 1	2	2	2	l									2	1	
CO2	2	2	2	I									2	I	
C03	2	2	2	1									2	1	
CO 4	2	2	2	I									2	2	
CO5	2	2	1	1									2	1	
CO6	2	1	1	1									2	2	

UNITI

Introduction to Linear Data Structures: Definition and importance of linear data structures, Abstract data types (ADTs) and their implementation, Overview of time and space complexity analysis for linear data structures. Searching Techniques: Linear & Binary Search, Sorting Techniques: Bubble soi1, Selection sort, Insertion Sort

UNITII

Linked Lists: Singly linked lists: representation and operations, doubly linked lists and circular linked lists, Comparing arrays and linked lists, Applications of linked lists

UNITIII

Stacks: Introduction to stacks: properties and operations, implementing stacks using arrays and Applications of stacks

UNIT IV

Queues: Introduction to queues: properties and operations, implementing queues using arrays and linked lists, Applications of queues in breadth-first search, scheduling, etc.

Deques: Introduction to deques (double-ended queues), Operations on deques and their applications.

UNIT IV

Graphs: Introduction, Graphs technology, Directed Graphs and Representation of Graphs
Trees: Introduction to Trees, Binary search Tree- Insertion, Deletion &Traversal

Hashing: Brief introduction to hashing and hash functions, Collision resolution techniques: chaining and open addressing, Hash tables: basic implementation and operations, Applications of hashing in unique identifier generation, caching, etc.

Textbooks:

1. Data Structures and algorithm analysis in C, Mark Allen W eiss, Pearson,2nd Edition.
2. Fundamentals of data structures in C, Ellis Horowitz, SartajSahni, Susan Anderson- Freed, Silicon Press, 2008

Reference Books:

1. Algorithms and Data Structures: The Basic Toolbox by Kurt Mehlhorn and Peter Sanders
2. C Data Structures and Algorithms by Alfred V. Aho, Jeffrey D. Ullman, and John E. Hopcroft
3. Problem Solving with Algorithms and Data Structures "by Brad Miller and David Ranum
4. Introduction to Algorithms by Thomas **H.** Cormen, Charles E. Leiserson, Ronald L. Rivest, and Clifford ·Stein
5. Algorithms in C, Parts 1-5 (Bundle): Fundamentals, Data Structures, Sorting, Searching, and Graph Algorithms "by Robert Sedgewick